Project Planning Phase

Product Backlog, Sprint Planning, Stories, & Story Points

Metadata

- **Date:** June 19, 2026
- **Team ID:** ------------------
- **Project Name:** AI-Enhanced Intrusion Detection System
- **Maximum Marks:** 8 Marks

1. Product Backlog, Sprint Schedule, and Estimation

This backlog lists the complete engineering tasks required to build the application features, estimated in agile story points (reflecting relative complexity and effort) and distributed across optimized execution sprints.

1. Sprint Backlog (Easy Explanation)

Sprint	Requirement	User Story No.	Simple Task	Points	Priority	Team Member
Sprint 1	Core Setup	USN-1	Create project folders, install libraries, and set up files	2	High	Python Developer
Sprint 1	ML Model Training	USN-2	Train the model using NSL-KDD dataset with Random Forest	5	High	ML Engineer
Sprint 1	Flask Server Setup	USN-3	Create main Flask app and page routing	3	High	Backend Developer
Sprint 1	UI Design	USN-4	Create Home and About pages with CSS styling	3	Medium	Frontend Developer
Sprint 2	Dashboard	USN-5	Build dashboard for traffic charts using Chart.js	5	High	Frontend Developer
Sprint 2	Prediction Form	USN-6	Create input form for live intrusion detection	5	High	Full Stack Developer
Sprint 3	Demo Buttons	USN-7	Add buttons for testing attack samples automatically	2	Medium	Frontend Developer
Sprint 3	Alert Logs	USN-8	Store alerts temporarily and display in table format	4	High	Backend Developer
Sprint 4	REST API	USN-9	Create API to accept JSON and return prediction results	4	Medium	Backend Developer
Sprint 4	Testing & Docs	USN-10	Test the system and prepare final documentation	2	Medium	DevOps / Lead

2. Project Tracker (Easy Explanation)

Sprint	Total Points	Duration	Start Date	End Date	Work Completed	Release Date
Sprint 1	13	6 Days	22 June 2026	27 June 2026	13	27 June 2026
Sprint 2	10	6 Days	29 June 2026	04 July 2026	10	04 July 2026
Sprint 3	6	6 Days	06 July 2026	11 July 2026	6	11 July 2026
Sprint 4	6	6 Days	13 July 2026	18 July 2026	6	18 July 2026

3. Velocity (Simple)

Velocity means **how much work the team finishes in one sprint.**

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Formula:

Velocity = Total Story Points Completed / Number of Sprints

Total Completed = **13 + 10 + 6 + 6 = 35**

Number of Sprints = **4**

Velocity = 35 ÷ 4 = 8.75 ≈ 9 points per sprint

✅ Team average speed = **9 story points per sprint**

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4. Burndown Chart (Simple Explanation)

Burndown chart shows how work decreases after every sprint.

Starting Work = **35 Points**

- After Sprint 1 → Remaining = **22**
- After Sprint 2 → Remaining = **12**
- After Sprint 3 → Remaining = **6**
- After Sprint 4 → Remaining = **0**

Chart Flow:

```
35 •
   |\
22 | \
   |  \
12 |   \
   |    \
 6 |     \
   |      \
 0 •-------•
   S1  S2  S3  S4
```

This means your project work is completed step-by-step until zero pending tasks.

Simple Summary:

- **Sprint 1** = Setup + Model + Flask + UI
- **Sprint 2** = Dashboard + Prediction Form
- **Sprint 3** = Demo Buttons + Alert Logs
- **Sprint 4** = API + Testing + Documentation

This is the easy version you can directly write in your project report.